S2CMamba: Structure-to-Contrast Semantic Mamba for Hyperspectral and LiDAR Joint Classification

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***Abstract*—Hyperspectral images (HSI) and light detection and ranging (LiDAR) provide complementary spectral and elevation information, making their fusion an effective solution for multimodal remote sensing (RS) classification. However, existing fusion methods often fail to adequately capture complex spatial topology and long-range contextual dependencies. Moreover, foreground–background semantic entanglement and the lack of semantic-guided fusion mechanisms further degrade feature discrimination. To address these challenges, a novel structure-to-contrast semantic Mamba (S2CMamba) network is proposed for the joint classification of HSI and LiDAR data. Specifically, a structure-aware state space module (S2SSM) module is designed for both HSI and LiDAR branches to encode multidirectional spatial structures and capture long-range global spatial dependencies within each modality. Then, the proposed semantic decoupling unit (SDU) explicitly separates foreground and background semantic responses in the fused HSI–LiDAR representations, which effectively suppresses background interference and alleviates semantic ambiguity during multimodal feature fusion. Based on the decoupled semantic priors, a contrast-guided feature aggregation module (CGAM) is developed to enable contrast-modulated local window aggregation over the fused HSI–LiDAR features, thereby improving intra-class compactness and enhancing class-boundary discrimination. Through experiments conducted on three commonly used HSI and LiDAR datasets, the performance of S2CMamba was consistently demonstrated to surpass that of state-of-the-art (SOTA) methods for HSI and LiDAR.**

***Index Terms*—Contrast-guided feature aggregation module (CGAM), hyperspectral image classification (HSIC), hyperspectral images–light detection and ranging (HSI–LiDAR) fusion, semantic decoupling unit (SDU), structure-aware state space module (S2SSM).**

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I. Introduction

THE rapid advancement of Earth observation technology in recent years, remote sensing (RS) [1], [2], [3], [4] imagery has been extensively applied for geographic information acquisition, environmental monitoring, and related fields [5], [6], [7], [8]. Hyperspectral images (HSI) simultaneously capture spatial information and rich spectral detail [9], [10], rendering them highly effective for material identification and land cover classification [11], [12], [13], [14]. However, HSIs predominantly depict 2-D spatial configurations and spectral attributes, constraining their capacity to differentiate objects with analogous spectral responses yet disparate geometric forms, such as asphalt roadways, sidewalks, bushes, and dwarf trees.

Light detection and ranging (LiDAR) data offers further 3-D structure details, elevation, and surface texture, effectively overcoming the spatial limitations of HSI [15], [16], [17]. LiDAR explicitly encodes height and geometric structure, providing reliable cues for spatial layout modeling and helping distinguish objects with similar spectral responses but different shapes. HSI captures fine-grained spectral signatures across materials, improving semantic discrimination between classes with similar geometry but different compositions. Consequently, the integration of HSI and LiDAR data significantly enhances classification accuracy, becoming a central research topic in multimodal RS analysis [18], [19].

Given that HSI and LiDAR data possess complementary characteristics, methods that combine analysis of these two data attract increased interest in the field of RS. To make effective use of such complementary approaches, various deep learning (DL) frameworks have been developed, based on convolutional neural networks (CNNs), Transformers, state-space models (SSMs), and hybrid variants of these.

CNN-based approaches have been proven to deliver outstanding performance in multimodal spatial-spectral feature extraction. Xu et al. [20] suggested a two-stage CNN architecture consisting of a dual-channel CNN and a cascade CNN. The former derives spatial-spectral information from HSI data, while the latter obtains elevation characteristics from LiDAR data, thereby improving classification efficacy. Hang et al. [21] established a hybrid CNN architecture in

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which two parallel CNNs extract spectral and spatial characteristics from HSI data and elevation data from LiDAR, respectively. This model attains superior classification accuracy by adeptly using complementary multimodal information via parameter sharing within neural layers and an integrated fusion strategy for features and decisions. Wang et al. [22] proposed S3F2Net to integrate spatial, spectral, and structural information, thereby improving classification performance in nonuniform regions with indistinct boundaries. Furthermore, MS2CANet [23] implemented multiscale and cross-modal attention mechanisms to capture contextual dependencies and augment the discriminative capacity of features. Despite these advances, CNN-based methods often struggle to capture extensive contextual dependencies and may exhibit oversmoothing effects in complex scenes.

Transformer-based approaches employ a self-attention mechanism to capture long-range contextual relationships and enhance global feature representations. The collaborative fusion network proposed by Tang et al. [24] demonstrates excellent performance across various land cover scenarios by integrating HSI and LiDAR features. Wang et al. [25] developed a hierarchical densely connected multiattention fusion network to capture multiscale spatial and spectral correlations and mitigate the impact of noisy or blurry pixels. Lu et al. [26] proposed a coupled adversarial learning framework to improve feature consistency across modalities and enhance generalization capabilities in heterogeneous regions. Furthermore, SpectralFormer [27] and the multimodal fusion transformer [28], [29] have enhanced classification accuracy by explicitly modeling long-range dependencies and intermodality interactions. The quadratic computational complexity of Transformers constrains scalability with high-resolution data and may lead to insufficient preservation of fine spatial details.

Over the past few years, SSM-based frameworks, particularly Mamba, have established their identity as a computationally efficient alternative with powerful global modeling capabilities. Ding et al. [30] proposed a global-local transformer network, which captures global contextual dependencies by incorporating state-space elements and reducing computing costs. Roy et al. [31] presented the spectral-space morphological attention Transformer, which encodes structural patterns in HSI data and improves feature representation. In addition, HLMamba [32] captures long-distance relationships in modalities by integrating edge contour information into the multimodal Mamba model and improves feature combination and classification efficiency. Despite these advances, existing Mamba-based approaches show limitations in structural identification capabilities across different domains and persistent semantic ambiguity, blurring class boundaries and reducing generalization potential.

Despite rapid advances in methods based on CNNs, Transformers, and Mamba, several critical challenges remain in HSI–LiDAR multimodal classification. Firstly, the intricate spatial structures and long-range contextual dependencies in HSI–LiDAR data remain inadequately modeled in a topologically consistent manner, particularly pronounced in scenarios featuring heterogeneous land cover distributions. Secondly, semantic information from foreground and background regions frequently becomes entangled during HSI–LiDAR feature fusion, diminishing feature discriminative power and reducing inter-class separability. Thirdly, existing HSI–LiDAR fusion strategies lack adaptive mechanisms to optimize multimodal representations under semantic guidance, resulting in limited robustness in boundary-sensitive regions.

To alleviate semantic ambiguity in HSI–LiDAR feature fusion and overcome the lack of adaptive mechanisms for semantically guided representation optimization in existing HSI–LiDAR fusion strategies, ConDSeg [33] proposes a contrast-driven segmentation method in medical image analysis that exploits semantic contrast cues to guide feature aggregation. Structural modeling preserves spatial continuity and topological consistency, which helps reveal transitions between heterogeneous regions and makes semantic contrast more distinguishable. Based on this observation, the proposed structure-to-contrast paradigm uses structurally consistent features to guide contrast-aware semantic refinement for more discriminative representations. Inspired by this idea, a structure-to-contrast semantic Mamba (S2CMamba) framework is developed for the joint classification of HSI and LiDAR data. The framework includes three key components: 1) a structure-aware state space module (S2SSM); 2) a semantic decoupling unit (SDU); and 3) a contrast-guided feature aggregation module (CGAM). Within the framework, the S2SSM is designed to achieve topologically consistent spatial modeling in both HSI and LiDAR feature branches. It captures multidirectional spatial continuity and long-range dependencies using a 2-D selective scan (SS2D) and structure-aware visual state space (SAVSS) layers, resulting in structurally consistent feature representations. To mitigate semantic ambiguity in multimodal representations, the SDU is proposed to explicitly separate foreground and background semantics through continuous semantic response modeling in fused HSI–LiDAR features. Based on the resulting semantic priors, the CGAM is further developed to conduct foreground–background contrast-modulated local window aggregation, where semantic-guided attention maps recalibrate neighborhood features to suppress background interference and strengthen class-boundary discrimination.

The key contributions of this article are listed as follows.

1) In this article, a novel S2CMamba framework is proposed for the joint classification of HSI and LiDAR data. The framework captures spatial topology and semantic contrast relationships in HSI–LiDAR data through structure-aware state space modeling and semantic-aware representation learning, thereby improving structural consistency and discriminative capability for multimodal feature fusion.
2) The S2SSM is utilized for the first time to encode multidirectional spatial structural features within each modality of HSI and LiDAR through the SS2D strategy and SAVSS layer, enabling enhanced modeling of long-range global spatial dependencies.
3) The SDU is designed for the first time to explicitly disentangle foreground and background semantics in fused HSI–LiDAR representations. Through continuous

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semantic response modeling under weak supervision derived from category labels, SDU generates stable semantic priors that effectively suppress cross-modal ambiguity and reduce irrelevant background interference.

4) Based on the semantic priors generated by SDU, the CGAM is used for the first time to enable HSI and LiDAR foreground-background contrast-modulated local window aggregation, thereby suppressing background co-occurrence interference and enhancing class-boundary discrimination in HSI and LiDAR. By using semantic attention maps to recalibrate neighborhood features while improving classification robustness and stability.

The subsequent sections of this article are organized as follows. Section II clearly examines related work. Section III describes the methodology. Section IV indicates experimental results obtained from three widely used datasets, and Section V summarizes this article's conclusion.

II. Related Work

A. Selective State Space Models

SSMs constitute a classical family of sequence modeling approaches, which describe the dynamic evolution of input sequences through recursive hidden-state transitions. Early SSM research was primarily represented by Kalman filtering models and was widely applied in signal processing and control systems [34]. Recently, with the introduction of structured state space models (S4), SSMs have been re-integrated into DL frameworks and demonstrated remarkable efficiency and stability in long-sequence modeling tasks. Under a discrete-time formulation, the S4 model can be expressed as

$$h_t = \mathbf{A}h_{t-1} + \mathbf{B}x_t, \quad \mathbf{y}_t = \mathbf{C}h_t \tag{1}$$

where x_t denotes the input sequence, h_t represents the hidden state, and $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are learnable state transition matrices. By recursively expanding the hidden states, the output can be expressed as a weighted sum over historical inputs $\mathbf{y}_t = \sum_{i=0}^{t} \mathbf{C}\mathbf{A}^{t-i}\mathbf{B}\mathbf{x}_i$, where i indexes the time steps of historical inputs. This form can be further written as a convolution over the input sequence $\mathbf{y} = \mathbf{K} * \mathbf{x}$, where the convolution kernel is defined as $\mathbf{K} = \{\mathbf{K}_i\}$ with $\mathbf{K}_i = \mathbf{C}\mathbf{A}^i\mathbf{B}$. This equivalence reveals the connection between the recurrent formulation and its convolutional implementation, allowing efficient parallel computation via structured convolution. The S4 efficiently models long-range dependencies by converting the recurrent formulation into a convolutional form. However, since S4 assumes a linear time-invariant (LTI) state transition matrix across the entire sequence, its capability to capture local key patterns and position-dependent information in complex scenarios is inherently limited, particularly in highly structured tasks such as vision and RS.

To overcome these limitations, the Mamba architecture introduces the selective state space model (S6) [35]. Unlike S4, S6 incorporates an input-dependent selective mechanism, allowing the state transition process to adapt dynamically according to the current input content. The discretized state update can be formulated as

$$h_t = \mathbf{A}(x_t)\, h_{t-1} + \mathbf{B}(x_t)\, x_t, \quad y_t = \mathbf{C}h_t \tag{2}$$

where the state transition matrix $\mathbf{A}(\mathbf{x}_t)$ and input projection matrix $\mathbf{B}(\mathbf{x}_t)$ are conditioned on the current input. This design enables content-aware and position-adaptive modeling. In practice, S6 is implemented via a selective scan mechanism. Although it can be viewed as a generalized convolutional process, the corresponding kernels are input-dependent and vary across positions, thereby breaking the standard LTI assumption. As illustrated in Fig. 1, the Mamba block adopts a structured design that integrates input projection, local convolution, nonlinear activation, and the S6 operator, forming a structured pipeline for selective information propagation and dynamic state updating. This design enhances the SSM's ability to capture input-dependent and position-dependent patterns while maintaining linear computational complexity. Benefiting from this formulation, Mamba maintains linear time complexity while effectively modeling long-range dependencies, achieving strong performance in natural language processing, speech modeling, and genomic sequence analysis.

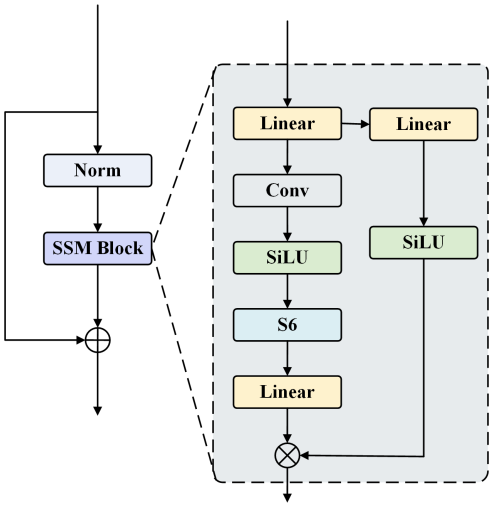


Fig. 1. Structure of mamba.

Following its success in sequential modeling, Mamba has been extended to visual tasks, leading to the development of visual state space (VSS) architectures. For instance, vision Mamba (ViM) achieves performance comparable to vision Transformers (ViTs) without relying on self-attention [36]; VMamba [37] further balances computational efficiency and representation capability; and plain-Mamba [38] employs fixed-width hierarchical stacking, demonstrating strong results in object detection and instance segmentation. These studies suggest that SSM-based visual modeling offers significant advantages in computational efficiency and global dependency modeling.

However, most existing VSS modules adopt fixed or simplified scanning strategies that primarily emphasize sequential dependency modeling while insufficiently preserving spatial structural continuity and direction-sensitive patterns. Given that different visual tasks require varying degrees of

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long-range dependency and fine-grained structural modeling, generic VSS designs are not directly suitable for complex RS scenarios, especially in images containing oblique structures, elongated objects, and intricate topological relationships. Therefore, task-oriented optimization of scanning strategies and structure-aware modeling within selective state space frameworks remains necessary for multimodal RS applications.

B. Attention Mechanisms

Transformer architectures and their self-attention mechanisms were originally developed for natural language processing and subsequently extended to vision tasks, forming the ViT and its variants. Self-attention explicitly models pairwise relationships across arbitrary positions, providing a powerful mechanism for capturing global contextual dependencies and thus gaining widespread adoption in RS image analysis.

In the standard Transformer framework, input features are linearly projected into query ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$) representations. The self-attention operation is defined as

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d}}\right)\mathbf{V} \tag{3}$$

where $\mathbf{K}^{\top}$ denotes the transpose of the key matrix, which enables matrix multiplication with the query matrix, and $\mathbf{Q}$ to compute pairwise similarity scores between sequence elements. d represents the feature dimensionality. This formulation enables adaptive global feature aggregation across all positions, effectively modeling long-range dependencies.

In RS, Transformers have been widely applied to HSI classification, semantic segmentation, and multimodal data fusion. Multihead attention (MHA) augments representational capacity by acquiring parallel projections across various subspaces

$$\begin{aligned}\text{MHA}(\mathbf{Q},\mathbf{K},\mathbf{V}) &= \text{Concat}(\text{head}_1,\ldots,\text{head}_H)\,\mathbf{W}^{O}\\ \text{head}_i &= \text{Attention}\left(\mathbf{Q}\mathbf{W}_i^{Q},\mathbf{K}\mathbf{W}_i^{K},\mathbf{V}\mathbf{W}_i^{V}\right)\end{aligned} \tag{4}$$

where H denotes the number of attention heads. For the ith head, $\mathbf{W}_i^{Q}$, $\mathbf{W}_i^{K}$, and $\mathbf{W}_i^{V}$ are the learnable projection matrices for query, key, and value, respectively, while $\mathbf{W}^{O}$ is the output projection matrix that maps the concatenated outputs of all heads to the final representation. This design promotes feature interaction between spectral and structural representations. For multimodal RS tasks, cross-attention (CA) mechanisms have also been introduced to explicitly model inter-modal interactions

$$\text{CA}\left(\mathbf{Q}^{(a)},\mathbf{K}^{(b)},\mathbf{V}^{(b)}\right) = \text{Softmax}\left(\frac{\mathbf{Q}^{(a)}\mathbf{K}^{(b)\top}}{\sqrt{d}}\right)\mathbf{V}^{(b)} \tag{5}$$

where superscripts a and b denote different modalities, CA facilitates intermodal alignment and complementary information exchange, improving multimodal feature representation.

Considering their excellent modeling capabilities, Transformer-based methodologies have inherent limits in the practical RS scenarios. The computational cost of self-attention increases quadratically with sequence length $O(N^2)$, leading to considerable computing and memory demands when handling high-resolution images or extensive geographical areas. Secondly, global attention processes often consolidate superfluous background information, especially in situations characterized by class imbalance or predominant background regions, which may undermine the discriminative representation of objectives.

To mitigate these issues, local attention, window-based attention, and multiscale attention mechanisms have been proposed to restrict the attention scope and reduce computational cost. However, these methods still rely on explicit similarity-based attention weighting. In multimodal deep fusion scenarios, object-related semantics and background context are often highly entangled in feature space, and existing attention mechanisms generally lack explicit constraints for foreground–background separation, limiting their ability to suppress contextual interference effectively.

In summary, although attention mechanisms effectively model feature dependencies via similarity-based weighting, their core formulation depends on explicit attention computation and global or local feature interaction. In complex RS scenes, multimodal fusion features typically contain both object-relevant semantics and substantial background context. Pure similarity-based attention allocation struggles to explicitly distinguish foreground and background semantics, leading to dispersed semantic responses and contextual interference. Introducing foreground–background semantic contrast priors into the feature aggregation process may therefore enhance discriminative representation while preserving computational efficiency.

III. Proposed Method

A. Architecture Overview

This section offers a comprehensive overview of the proposed S2CMamba framework, with its general design illustrated in Fig. 2(a). The network contains three key components:1) the S2SSM module; 2) the SDU module; and 3) the CGAM module, allowing effective joint classification of HSI and LiDAR data.

Given the HSI data $\mathbf{X}_h \in \mathbb{R}^{H\times W\times L}$ and LiDAR data $\mathbf{X}_l \in \mathbb{R}^{H\times W\times 1}$, where H and W denote the height and width of the image, respectively, and L represents the spectral dimension of the HSI image, a local patch centered at each pixel is extracted from each modality. The resulting patches are denoted by $\mathbf{P}_h \in \mathbb{R}^{p\times p\times L}$ and $\mathbf{P}_l \in \mathbb{R}^{p\times p\times 1}$, respectively.

Subsequently, the HSI and LiDAR patches are input into shallow embedding modules consisting of 3×3 convolution, batch normalization (BN), and ReLU activation. These modules extract preliminary spectral–spatial features and structural-spatial features while preserving spatial resolution, thereby achieving an initial alignment between heterogeneous modality representations. The lightweight gated bottleneck convolution (GBC) operator is embedded within both the shallow embedding module and subsequent structure-aware modeling stages to reduce computational complexity while maintaining discriminative capability. Building upon this foundation, the S2SSM module is designed. It leverages the SS2D together with SAVSSLayer to capture multidirectional spatial

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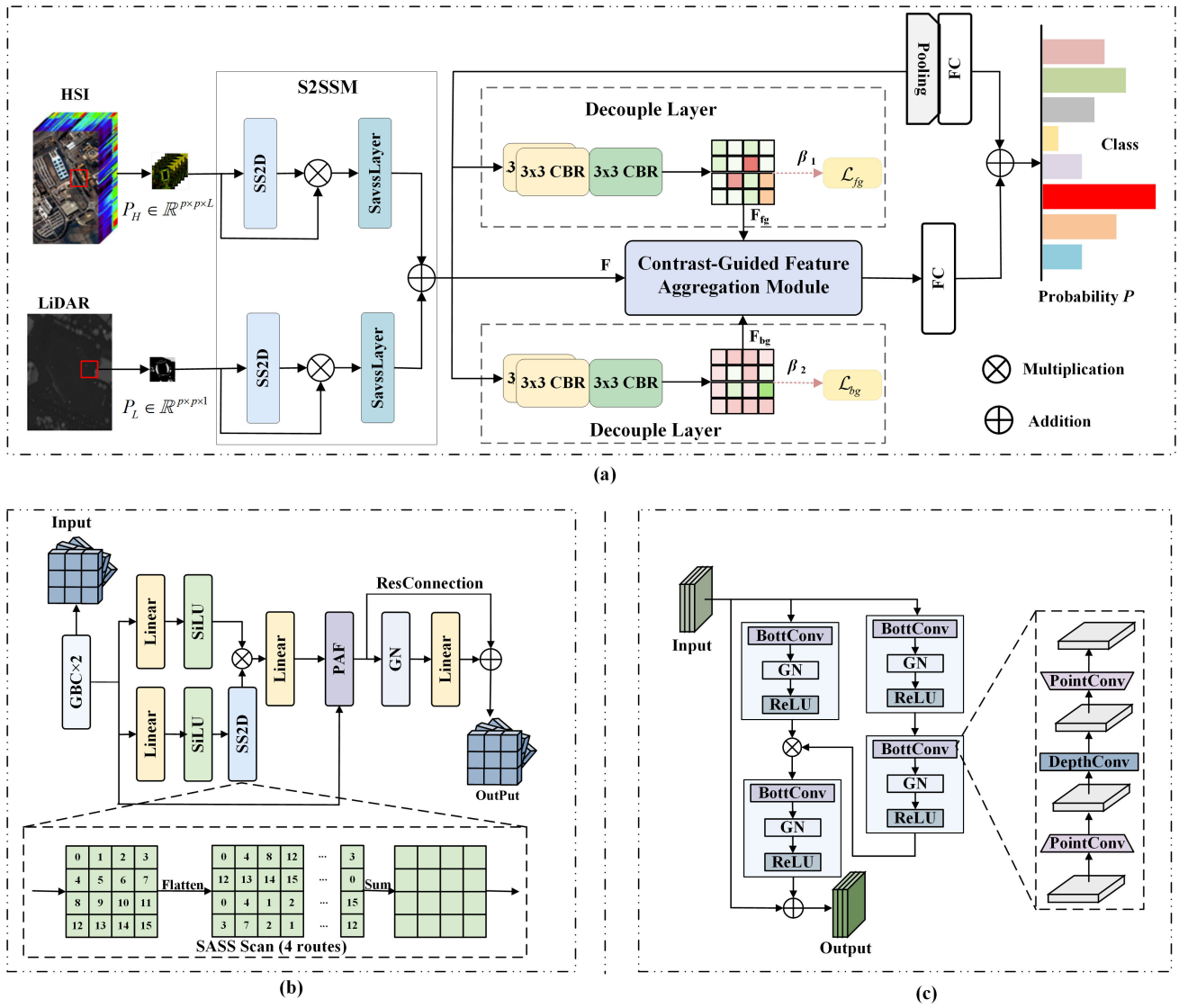


Fig. 2. (a) Outlines the overall structure of the S2CMamba. (b) Illustrates the structure of the S2SSM block. (c) Architecture of GBC, where GN denotes group normalization.

continuity and structural correlations between HSI and LiDAR features. By modeling these spatial dependencies through structured state space representations, the module enables effective multimodal structural interaction.

After the S2SSM module, the obtained joint features $\mathbf{F}$ encode rich multidirectional structural dependencies and spatial topology. These structurally consistent features are subsequently fed into the SDU for semantic decoupling, while also implicitly providing spatial coherence constraints for the following CGAM. This module explicitly separates foreground and background semantics at the feature representation level and suppresses background interference via dynamically weighted semantic constraints. Consequently, it alleviates semantic confusion and soft-boundary issues commonly observed in multimodal fusion, providing clearer and more stable semantic priors for subsequent feature aggregation. Furthermore, the CGAM module is proposed, which leverages the foreground and background features $\mathbf{F}_{fg}, \mathbf{F}_{bg}$ generated by the SDU module as contrastive guidance for feature refinement. The joint features $\mathbf{F}$ are adaptively reweighted and aggregated to enhance discriminative representation. By employing a local window attention mechanism, CGAM suppresses background co-occurrence interference while strengthening the discriminability of foreground regions, achieving discriminative multimodal feature refinement.

Finally, the aggregated features $\hat{\mathbf{F}}$ are sent into a classifier to produce land-cover category predictions. During training, the network is optimized using a multitask objective that jointly considers the classification loss and the semantic decoupling loss introduced by the SDU module. The classification loss supervises the final land-cover prediction, while the semantic decoupling loss constrains the learning of foreground and background semantic responses, thereby encouraging more discriminative multimodal feature representations. Through the progressive collaboration of the S2SSM, SDU, and CGAM modules under the joint training objective, the proposed

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S2CMamba learns robust and discriminative representations for complex RS scenes.

B. Lightweight GBC

Considering the high dimensionality and computational complexity of HSI and LiDAR joint features, a lightweight GBC module is adopted as the fundamental feature modeling unit. GBC integrates low-rank bottleneck modeling with a gating mechanism to significantly reduce parameter size and computational cost while enhancing the representation capability for fine-grained structural information, thereby providing efficient feature representations for subsequent structure-aware modeling.

Specifically, GBC is built upon bottleneck convolution (BottConv) [39], which approximates standard convolution using a low-rank decomposition. For a standard convolution with spatial size $p \times p$ and the input is $\mathbf{x}$, the linear response can be expressed as

$$\mathbf{Y} = \mathbf{W}_{\text{conv}}\mathbf{x} + b \tag{6}$$

where $\mathbf{W}_{\text{conv}} \in \mathbb{R}^{f_o \times (p^2 d_i)}$ denotes the convolution matrix, while f_o and d_i are the numbers of output and input channels, respectively, and b denotes the bias. Assuming that $\mathbf{Y}$ lies in a low-rank subspace with rank f_r ($f_r \ll f_o$), a low-rank factorization is introduced to decompose the convolution weight as $\mathbf{W}_{\text{conv}} = \mathbf{U}\mathbf{W}_r^\top$, where $\mathbf{W}_r \in \mathbb{R}^{(p^2 d_i) \times f_r}$ projects the high-dimensional input features into a low-rank subspace and $\mathbf{U} \in \mathbb{R}^{f_o \times f_r}$ maps the low-rank features back to the original output channel space. Accordingly, the BottConv response can be simplified as

$$\mathbf{Y} = \mathbf{U}\mathbf{W}_r^\top \mathbf{x} + b'. \tag{7}$$

In BottConv, pointwise convolutions are used for projection to and reconstruction from a low-rank space, reducing computational cost. Depthwise convolution is then applied in the reduced space to capture spatial information with little cost. Consequently, GBC implements a gating mechanism [40], [41] to facilitate the adaptive modulation of joint features. As illustrated in Fig. 2(c), the input feature $\mathbf{x} \in \mathbb{R}^{H \times W \times D}$ is entirely retained as a residual branch $\mathbf{x}_{\text{residual}} = \mathbf{x}$, providing a foundation for subsequent residual connections and ensuring that core feature information is not lost. The input feature $\mathbf{x}$ passes through BottConv and normalization, then through ReLU activation to obtain the main feature $\mathbf{x}_1$ and the gating feature $g_2(\mathbf{x})$

$$\begin{aligned} g_1(\mathbf{x}) &= \text{ReLU}\left(\text{Norm}_1\left(f_1(\mathbf{x})\right)\right) \\ \mathbf{x}_1 &= \text{ReLU}\left(\text{Norm}_2\left(\text{BottConv}_2\left(g_1(\mathbf{x})\right)\right)\right) \\ g_2(\mathbf{x}) &= \text{ReLU}\left(\text{Norm}_3\left(\text{BottConv}_3(\mathbf{x})\right)\right). \end{aligned} \tag{8}$$

Then $\mathbf{x}_1$ and $g_2(\mathbf{x})$ are combined through a Hadamard product for element-wise weighting to form the gating map $\mathbf{M}(\mathbf{x})$. This operation adaptively reweights feature responses, enhancing informative details while suppressing noise

$$\mathbf{M}(\mathbf{x}) = \mathbf{x}_1 \odot g_2(\mathbf{x}). \tag{9}$$

The gated feature $\mathbf{M}(\mathbf{x})$ is further transformed by BottConv, normalization, and activation to produce $\mathbf{y}$, which is then combined with $\mathbf{x}_{\text{residual}}$ for the final output

$$\begin{aligned} \mathbf{y} &= \text{ReLU}\left(\text{Norm}_4\left(\text{BottConv}_4\left(\mathbf{M}(\mathbf{x})\right)\right)\right) \\ \mathbf{y}_{\text{out}} &= \mathbf{y} + \mathbf{x}_{\text{residual}}. \end{aligned} \tag{10}$$

This gating design is particularly suitable for HSI-LiDAR joint modeling scenarios, enabling adaptive balancing of the contributions of spectral information and structural information according to different ground object categories and spatial locations.

C. Structure-Aware State Space Module

As shown in Fig. 2(b), the S2SSM module is located in the structural modeling stage of the overall framework and aims to perform multidirectional structural enhancement and global dependency modeling on the shallow-embedded HSI and LiDAR features. Conventional visual Mamba networks typically adopt fixed scanning strategies, such as raster, serpentine, or bidirectional scans. Although these strategies maintain horizontal or vertical semantic continuity, they are insufficient for capturing complex topological structures, especially in scenes containing oblique objects, elongated targets, and irregular spatial patterns.

To address these limitations in HSI and LiDAR scenarios, a structure-aware scanning strategy (SASS) is proposed, as shown in Fig. 3. Specifically, SASS consists of four scanning paths, including two parallel serpentine paths and two diagonal serpentine paths. By introducing diagonal serpentine traversal, multidirectional adjacency relationships are constructed, enabling semantic continuity in regular regions while preserving structural completeness in oblique areas. Consequently, SASS provides stable multidirectional structural priors for subsequent state space modeling.

After the HSI and LiDAR features are first extracted by a convolutional encoder, flattened, linearly embedded, and then undergo position encoding, they are input as sequences into the S2SSM. The processing equations are formulated as

$$\begin{aligned} \overline{\mathbf{P}} &= e^{\Delta \mathbf{P}} \\ \overline{\mathbf{Q}} &= (\Delta \mathbf{P})^{-1}\left(e^{\Delta \mathbf{P}} - \mathbf{I}\right) \cdot \Delta \mathbf{Q} \\ \mathbf{z}_k &= \overline{\mathbf{P}}\mathbf{z}_{k-1} + \overline{\mathbf{Q}}\mathbf{w}_k \\ \mathbf{u}_k &= \mathbf{R}\mathbf{z}_k + \mathbf{S}\mathbf{w}_k \end{aligned} \tag{11}$$

where the input $\mathbf{w} \in \mathbb{R}^{t \times D}$ is the sequence feature from either HSI or LiDAR branch, $\mathbf{P} \in \mathbb{R}^{G \times D}$ controls the hidden spatial-spectral state, $\mathbf{S} \in \mathbb{R}^{D \times D}$ is used to initialize the skip connection for input, $\mathbf{z}_k$ represents the specific hidden state at time step k, and $\mathbf{Q} \in \mathbb{R}^{G \times D}$ and $\mathbf{R} \in \mathbb{R}^{G \times D}$ are trainable matrices with hidden spatial dimensions G and spectral-spatial dimensions D, respectively. SS2D performs selective scanning along the multidirectional paths defined by SASS. $\mathbf{u}_k$ represents the output at time step k.

Since SS2D incorporates the multidirectional adjacency relationships established by SASS during scanning, the state transition matrix preserves comprehensive topological information across directions, allowing the hidden state to

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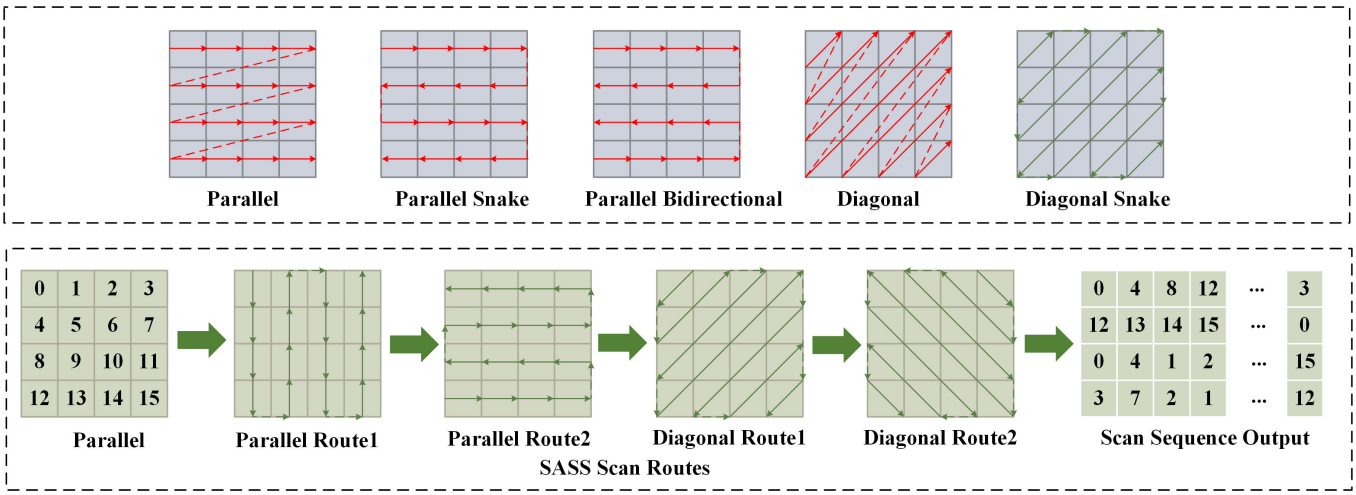


Fig. 3. Graphical representation of the proposed SASS and other scanning strategies. The first row shows the four common single-scan paths as well as the proposed diagonal snake path. The second row illustrates the sequence of the proposed SASS scanning strategy.

capture more complex spatial structures of HSI and LiDAR data. Therefore, the output features integrate multidirectional structural cues and cross-modal correlations, enhancing the representation of complex land-cover patterns.

To prevent the degradation of pixel-wise spectral–spatial correspondence during sequence aggregation, a pixel-aware fusion (PAF) mechanism is introduced within S2SSM. PAF fuses the original spatial features with SS2D-enhanced features using residual connections, thereby preserving fine-grained details and facilitating feature propagation. Meanwhile, GBC is embedded in the layer outputs for feature refinement, further improving discriminative capability in structurally complex regions.

D. Semantic Decoupling Unit

Due to the substantial differences between HSI and LiDAR data in imaging mechanisms and semantic representations, direct feature fusion often introduces redundant information irrelevant to object discrimination, leading to semantic ambiguity. To address this issue, the SDU is designed to explicitly separate object-related semantics from contextual background semantics based on the fused features generated by S2SSM.

Given the fused feature $\mathbf{F} \in \mathbb{R}^{B\times D\times H\times W}$, SDU employs two parallel branches composed of successive CBR blocks to decouple it into foreground-related and background-related semantic features, denoted as $\mathbf{F}_{fg}, \mathbf{F}_{bg} \in \mathbb{R}^{B\times D\times H\times W}$, respectively. To further obtain semantic confidence at each spatial location, the decoupled features are first compressed along the channel dimension and then normalized by a Sigmoid function to generate continuous foreground and background response maps

$$\begin{aligned}\mathbf{P}_{fg} &= \sigma\left(\frac{1}{D}\sum_{c=1}^{D}\mathbf{F}_{fg}^{(c)}\right)\\ \mathbf{P}_{bg} &= \sigma\left(\frac{1}{D}\sum_{c=1}^{D}\mathbf{F}_{bg}^{(c)}\right)\end{aligned} \tag{12}$$

where $\mathbf{P}_{fg}, \mathbf{P}_{bg} \in \mathbb{R}^{B\times H\times W}$ denote the foreground and background response maps, respectively, and $\sigma(\cdot)$ is the Sigmoid activation.

Instead of requiring pixel-level segmentation annotations, SDU constructs weak semantic supervision from the category labels and optimizes the response maps with binary cross-entropy (BCE) loss

$$\begin{aligned}\mathcal{L}_{\text{sdu}} &= \beta_1\mathcal{L}_{fg} + \beta_2\mathcal{L}_{bg}\\ \mathcal{L}_{fg} &= \text{BCE}\left(\mathbf{P}_{fg}, \mathbf{Y}_{fg}\right)\\ \mathcal{L}_{bg} &= \text{BCE}\left(\mathbf{P}_{bg}, \mathbf{Y}_{bg}\right)\end{aligned} \tag{13}$$

and $\mathbf{Y}_{fg}, \mathbf{Y}_{bg} \in \mathbb{R}^{B\times H\times W}$ are the expanded foreground/background supervisory maps. To alleviate the imbalance between foreground and background responses, adaptive weighting coefficients are introduced and computed from the mean activation intensity of the response maps

$$\begin{aligned}\beta_1 &= \frac{1}{\tanh\left(\frac{1}{N_p}\sum_{i=1}^{N_p}\mathbf{P}_{fg,i}\right)}\\ \beta_2 &= \frac{1}{\tanh\left(\frac{1}{N_p}\sum_{i=1}^{N_p}\mathbf{P}_{bg,i}\right)}\end{aligned} \tag{14}$$

where $N_p = H \times W$ denotes the number of spatial positions. The $\tanh(\cdot)$ function constrains the weights to the range $[1, +\infty)$, effectively avoiding numerical explosion caused by excessively small feature proportions, thereby ensuring the stability of the training process. Through the above semantic response modeling, SDU suppresses semantic mixing in the fused representation and provides foreground–background contrastive priors for subsequent feature aggregation. It is worth noting that the input feature $\mathbf{F}$ is generated by S2SSM with explicit structure-aware modeling. Therefore, the derived semantic priors $\mathbf{F}_{fg}$ and $\mathbf{F}_{bg}$ inherently preserve spatial continuity and topological consistency. This enables the SDU process to remain structurally aligned, which provides reliable and structurally consistent guidance for the subsequent contrast-driven aggregation in CGAM.

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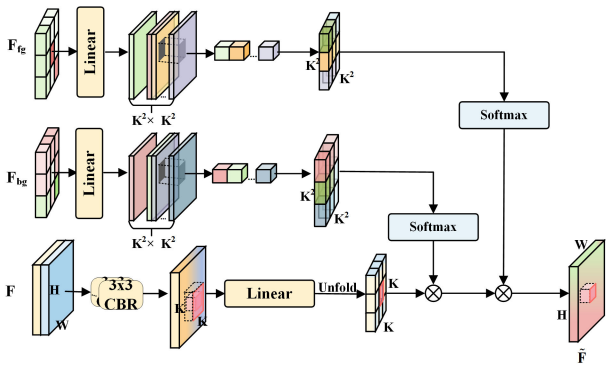


Fig. 4. Structure of CGAM.

E. Contrast-Guided Feature Aggregation Module

To further enhance discriminative representation under complex backgrounds, the CGAM module is designed to refine fused features under the guidance of foreground–background semantic contrast generated by SDU. As shown in Fig. 4, CGAM first employs multiple CBR modules for preliminary local fusion and feature enhancement. Let the refined feature after CBR processing be denoted as $\mathbf{F}' \in \mathbb{R}^{H\times W\times D}$. Subsequently, each feature vector at spatial position is projected into a value vector through a linear layer with learnable weight $\mathbf{W}_v \in \mathbb{R}^{D\times D}$, yielding the value feature map $\mathbf{V} \in \mathbb{R}^{H\times W\times D}$. Then, for each spatial location (i, j), a local neighborhood of size $K \times K$ centered at (i, j) is collected to capture local contextual dependencies. The corresponding neighborhood feature set is defined as

$$\mathbf{V}_{\Delta_{i,j}} = \left\{ \mathbf{V}_{i+p-\lfloor \frac{K}{2} \rfloor, j+q-\lfloor \frac{K}{2} \rfloor} \middle| \, 0 \le p, q < K \right\} \tag{15}$$

where $\mathbf{V}_{\Delta_{i,j}} \in \mathbb{R}^{D\times K^2}$. Next, semantic-guided attention maps for the foreground and background features are generated through linear projections

$$\mathbf{A}_{fg} = \mathbf{W}_{fg}\mathbf{F}_{fg} \qquad \mathbf{A}_{bg} = \mathbf{W}_{bg}\mathbf{F}_{bg} \tag{16}$$

where $\mathbf{W}_{fg}, \mathbf{W}_{bg} \in \mathbb{R}^{D\times K^4}$ are learnable projection matrices. Although the feature maps are generated from the SDU, it is important to note that $\mathbf{F}_{fg}$ and $\mathbf{F}_{bg}$ are derived from the structure-aware feature $\mathbf{F}$ produced by S2SSM. As a result, the generated attention implicitly inherits structural continuity and spatial topology, enabling the aggregation process to be guided by both semantic contrast and underlying structural consistency. For each spatial location (i, j), the projected attention responses are reshaped into two local attention matrices, $\hat{\mathbf{A}}_{fg_{i,j}}, \hat{\mathbf{A}}_{bg_{i,j}} \in \mathbb{R}^{K^2\times K^2}$, where $\hat{\mathbf{A}}_{fg_{i,j}}$ encodes the foreground-aware correlation structure and $\hat{\mathbf{A}}_{bg_{i,j}}$ represents the background-aware contextual dependency within the $K \times K$ neighborhood centered at (i, j). After Softmax normalization, the contrast-enhanced feature aggregation is formulated as

$$\hat{\mathbf{V}}_{\Delta_{i,j}} = \text{Softmax}\left(\hat{\mathbf{A}}_{fg_{i,j}}\right) \otimes \left(\text{Softmax}\left(\hat{\mathbf{A}}_{bg_{i,j}}\right) \otimes \mathbf{V}_{\Delta_{i,j}}\right) \tag{17}$$

where $\otimes$ denotes matrix multiplication. The aggregated feature is then obtained by summing the weighted features in the local window. Specifically, the aggregated feature at spatial position (i, j) can be formulated as

$$\hat{\mathbf{F}}_{i,j} = \sum_{0\le m,n<K} \hat{\mathbf{V}}^{i,j}_{\Delta_{i+m-\lfloor \frac{K}{2} \rfloor, j+n-\lfloor \frac{K}{2} \rfloor}}. \tag{18}$$

By incorporating semantic contrast priors, CGAM enhances the separability between foreground and background representations while suppressing contextual interference, thereby yielding more discriminative and stable features for classification. Overall, this article proposes a structure-contrast interaction mechanism. The structure-aware features generated by S2SSM guide semantic decoupling in SDU, and the resulting contrastive semantic prior further adjusts feature aggregation in CGAM. This progressive interaction enhances the discriminative capability and robustness of multimodal feature fusion.

F. Classification Stage

A lightweight linear classifier is employed in the classification stage to optimize computational efficiency and generalization. The enhanced feature $\hat{\mathbf{F}}$ and the original fused feature $\mathbf{F}$ are mapped to category logits via a dual-branch linear projection

$$\mathbf{X}_{cls} = f_{\text{linear}}\left(\hat{\mathbf{F}}\right) + f_{\text{linear}}\left(f_{\text{pooling}}\left(\mathbf{F}\right)\right) \tag{19}$$

where f_{linear} denotes a linear fully-connected (FC) layer, and f_{pooling} represents a global pooling operation applied to the original fused feature $\mathbf{F}$. The classification loss is formulated as the cross-entropy loss

$$\mathcal{L}_{cls} = -\frac{1}{N_s}\sum_{i=1}^{N_s}\sum_{c=1}^{C} y_{i,c}\log\left(p_{i,c}\right) \tag{20}$$

where N_s is the number of samples, C is the number of classes, and $y_{i,c}$ and $p_{i,c}$ denote the ground-truth label and predicted Softmax probability, respectively. The overall loss is defined as

$$\mathcal{L}_{total} = \mathcal{L}_{sdu} + \mathcal{L}_{cls} \tag{21}$$

where $\mathcal{L}_{sdu}$ is the semantic decoupling loss introduced by the SDU module. Note that $\mathcal{L}_{sdu}$ is only used during training to guide semantic distribution learning and is not involved in inference.

IV. Experiments

This section evaluates the proposed S2CMamba on three public HSI–LiDAR datasets, namely Houston2013, Trento, and MUUFL. We first introduce the experimental setup, including implementation details, training settings, and evaluation metrics. Comparisons with eight advanced methods are then conducted to assess the performance of S2CMamba. Finally, ablation studies are performed on S2SSM, SDU, CGAM, and the structural modeling design based on SS2D and SAVSSLayer. The results verify the effectiveness of the proposed components and demonstrate the superiority of S2CMamba in terms of classification accuracy and robustness.

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TABLE I
DETAILED CATEGORY INFORMATION FOR THREE DATASETS

No.	Houston			Trento			MUUFL		
	Class name	Train.	Test.	Class name	Train.	Test.	Class name	Train.	Test.
C1	Healthy grass	30	1221	Apple Trees	10	4024	Trees	20	23226
C2	Stressed grass	30	1224	Buildings	10	2893	Mostly Grass	20	4250
C3	Synthetic grass	30	667	Ground	10	469	Mixed Ground Surface	20	6862
C4	Trees	30	1214	Woods	10	9113	Dirt and Sand	20	1806
C5	Soil	30	1212	Vineyard	10	10491	Road	20	6667
C6	Water	30	295	Roads	10	3164	Water	20	446
C7	Residential	30	1238				Buildings Shadow	20	2213
C8	Commercial	30	1214				Buildings	20	6220
C9	Road	30	1222				Sidewalk	20	1365
C10	Highway	30	1197				Yellow Curb	20	163
C11	Railway	30	1205				Cloth Panels	20	249
C12	Parking Lot 1	30	1203						
C13	Parking Lot 2	30	439						
C14	Tennis Court	30	398						
C15	Running Track	30	630						
-	Total	450	14579	Total	60	30154	Total	220	53467

A. Datasets

The experiment utilized three common datasets: the Houston [42] dataset, the Trento [43] dataset, and MUUFL [44], [45] dataset. These three datasets will now be described in detail.

1) Houston: The dataset contains images captured by the combined spectral imager (CASI) and encompasses the University of Houston campus and its adjacent urban vicinity, which integrates HSI data with a laser-scanned digital surface model (DSM), with a spatial resolution of 345×1905 pixels and a spatial accuracy of 2.5 m. Specifically, the HSI comprises 144 spectral bands covering a wavelength range of 0.38 to 1.35 μm. Furthermore, the Houston dataset contains 15 029 classified samples representing 15 different land cover types.

2) Trento: The dataset was collected in rural areas around the city of Trento in southern Italy and consists of HSI and LiDAR datasets. The spatial dimensions are 166×600 pixels with a spatial resolution of 1 m. The HSI data covers the wavelength range of 0.42–0.99 μm and consists of 63 spectral bands. In addition, the LiDAR data was acquired using a 3100EA sensor. This dataset contains 30 214 labeled samples covering six types of land cover.

3) MUUFL: The dataset originates from the Gulf Harbor Campus of the University of Southern Mississippi in Long Beach, Mississippi, USA, and consists of HSI and LiDAR datasets. The HSI data has a spatial resolution of 325×220 pixels and comprises 64 spectral bands. Notably, the LiDAR data was acquired using an ALTM sensor employing a 1064 nm wavelength laser. This dataset comprises 53 687 annotated samples covering 11 land cover categories. Detailed classification information for the three datasets is presented in Table I.

B. Experimental Setup

1) Implementation Details: All experiments were conducted in PyTorch on a server equipped with an Intel Xeon E5-2696v4 CPU (2.20 GHz) and four NVIDIA Tesla V100 SXM2 GPUs with 16 GB memory. S2CMamba was trained using Adam with a learning rate of 0.003, a batch size of 64, and 500 epochs. The spatial sizes of the multiscale inputs were set to 8×8 and 16×16.

2) Comparison Methods: To validate the effectiveness of the proposed S2CMamba method, eight representative multimodal classification methods were selected for comparison, covering several mainstream categories of HSI and LiDAR fusion. Specifically, CNN-based frameworks (MICFNet [24], MS2CANet [23]) are included for their strong local feature extraction capability. Transformer-based frameworks (GLTNet [30], IDNet [46]) are adopted due to their advantage in modeling long-range dependencies. In addition, Mamba-based frameworks (HLMamba [32], E-Mamba [47]) are considered as emerging SSMs with efficient global modeling ability. Furthermore, to provide a more comprehensive evaluation, two representative hybrid and emerging architectures are also incorporated: 1) HyperMLP [48] and 2) S3F2Net [22]. The selection provides a comprehensive benchmark for evaluating local feature extraction, global dependency modeling, and cross-modal feature fusion capabilities.

1) *MICF-Net:* The CNN-based method employs a dual-branch CNN for preliminary feature extraction, followed by a dual-branch cross-modal attention fusion transformer (CMAFT) to capture global contextual dependencies. The adaptive mask modulation (AMM) balances cross-modal learning, whilst the semantic-guided feature fusion (SGFF) module facilitates mutually guided learning between HSI and LiDAR data.
2) *MS2CANet:* This method employs spatial-spectral cross-modal attention (S2CA) and effective feature recalibration (EFR) to extract and enhance multiscale features. The S2CA module facilitates interactive information exchange between HSI and LiDAR data. The CNN-based framework excels in complex regions with strong spatial-spectral variations.

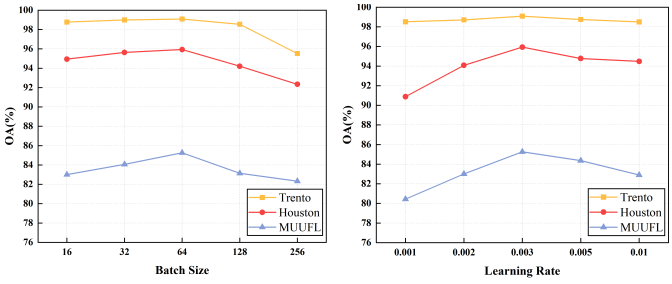
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3) *GLTNet:* This method is a Transformer-based multimodal fusion framework that combines multiscale local spatial feature learning with global spectral dependency modeling. Convolutional operations extract local spatial features, while a Transformer captures long-range spectral dependencies from HSI and LiDAR data. A probabilistic decision fusion strategy further enhances robustness, enabling effective exploitation of complementary local–global information across modalities.
4) *IDNet:* This method enhances detail representation while maintaining strength constraints, by embedding Laplacian high-frequency features into the Transformer. Transformer blocks with dynamic range awareness and spatial–spectral feature grouping capture local–global high-frequency features, while differential convolution emphasizes detailed information for adaptive multimodal fusion.
5) *HLMamba:* This method introduces a gradient joint algorithm for LiDAR edge extraction and a multimodal feature extraction module (MFEM) for HSI, LiDAR, and edge contour features. A multimodal Mamba fusion module (MMFM) models long-range dependencies and integrates intermodal features for improved classification.
6) *E-Mamba:* This method applies the Mamba architecture with a dual-branch siamese VSS block to extract features from HSI and LiDAR. The SS2D mechanism generates multiple image patch sequences processed by S6 blocks. Cross-Mamba feature extraction and modality-guided Mamba fusion blocks enhance cross-modal interaction and selective feature integration.
7) *HyperMLP:* Based on the MLP-mixer architecture, this method aims to address the issues of insufficient spatial feature construction and limited multimodal fusion capabilities in standard networks. Through unsupervised superpixel embedding, feature shuffling, and bilateral modulation strategies, the method enhances the feature's discriminative ability, diversity, and semantic alignment.
8) *S3F2Net:* This method employs a CNN with shared weights to extract local features while utilizing a graph convolutional network (GCN) to learn global topological structures. Dynamic node updating and multilevel information fusion enable effective integration of HSI and LiDAR data, enhancing classification accuracy and reliability.

3) Evaluation Metrics: Overall accuracy (OA), average accuracy (AA), and the kappa coefficient (Kappa) were adopted as evaluation metrics. All results are reported as the mean and standard deviation over ten independent runs, and the comparison method followed the same training approach adopted in this article to obtain the optimal parameters.

C. Parameter Analysis

1) Analysis of the Batch Size: The batch size affects both training stability and generalization performance. We evaluate the model using candidate values {16, 32, 64, 128, 256} while keeping other parameters fixed. As shown in Fig. 5(a), the OA on all three datasets increases as the batch size grows from 16 to 64, indicating that a moderate batch size leads to more stable gradient estimation and improved feature learning. When the batch size further increases to 128 and 256, the performance drops across all datasets. This suggests that overly large batch sizes reduce the stochasticity of optimization and weaken generalization. Overall, a batch size of 64 provides a good balance between training stability and classification accuracy.



(a) (b)

Fig. 5. Analysis of different parameters on OA across the Trento, Houston, and MUUFL datasets. (a) Effect of batch size. (b) Effect of learning rate.

2) Analysis of the Learning Rate: The learning rate is an important hyperparameter that influences convergence and final performance. We evaluate the model using candidate values {0.001, 0.002, 0.003, 0.005, 0.01}. As shown in Fig. 5(b), the OA on all datasets improves as the learning rate increases from 0.001 to 0.003, indicating that a slightly larger learning rate helps the model converge more effectively. When the learning rate further increases beyond 0.003, the performance starts to drop, with a more noticeable decline on the Houston dataset. This indicates that excessively high learning rates can reduce the stability of the optimization process. Overall, a learning rate of 0.003 achieves the best performance across the datasets.

D. Experimental Results and Analysis

To evaluate the proposed S2CMamba, comparative experiments were conducted on the Trento, Houston, and MUUFL datasets. The classification maps obtained by different methods are shown in Figs. 6–8. The corresponding quantitative results, including per-class accuracy, OA, AA, and Kappa, are reported in Tables II–IV, where the best values are highlighted in bold. Detailed analysis is provided below.

1) Houston Dataset: Table II summarizes the classification results on the Houston dataset. S2CMamba achieves the best performance, with OA, AA, and Kappa reaching 95.93%, 96.46%, and 95.59%, respectively. Compared with HLMamba, it improves these three metrics by 1.29%, 1.13%, and 1.39%, respectively, and also surpasses E-Mamba by 0.82%, 0.57%, and 0.88%. Existing HSI–LiDAR classification methods mainly differ in their feature modeling mechanisms. MICFNet and MS2CANet rely on convolutional operations and therefore focus more on local representations. HyperMLP and IDNet strengthen global interaction, but their ability to

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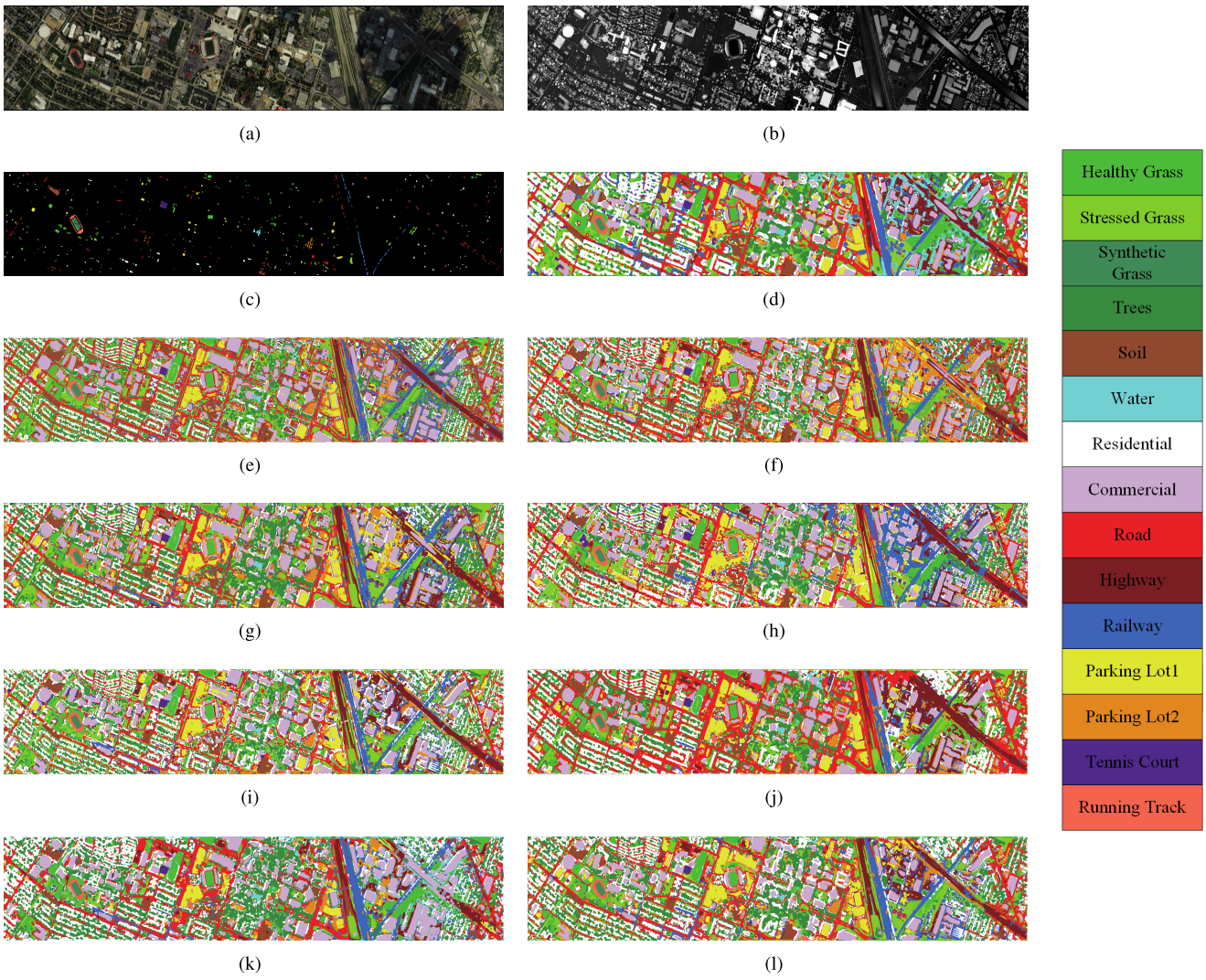


Fig. 6. Classification map of the various methods used in the Houston dataset. (a) Pseudocolor composite image based on bands 64, 43, and 20 of the HSI data. (b) Grayscale image for DSM based on LiDAR data. (c) Ground truth. (d) MICFNet (90.33%). (e) HyperMLP (93.93%). (f) GLTNet (94.29%). (g) MS2CANet (93.08%). (h) IDNet (94.35%). (i) HLMamba (94.64%). (j) E-Mamba (95.11%). (k) S3F2Net (84.86%).(l) Our methods (95.93%).

characterize structural patterns remains limited. S3F2Net introduces CNN and GCN to jointly model local features and global topology, improving multimodal fusion from different views. HLMamba and E-Mamba benefit from the linear-complexity property of Mamba for long-range dependency modeling, but they do not explicitly address structural consistency and semantic interference in multimodal fusion. S2CMamba alleviates these issues through S2SSM, SDU, and CGAM. S2SSM captures structural dependencies, SDU suppresses semantic confusion between foreground and background, and CGAM further improves feature aggregation. As shown in Table II, the proposed method achieves notable improvements on several categories. Significant improvements were observed in road (C9), highway (C10), and parking lot 1 (C12), where S2CMamba achieved the highest accuracy, indicating its advantage in handling categories with complex spatial structures. Although the performance was close to the optimal method for synthetic grass (C3), trees (C4), and parking lot 2 (C13), it was not always the highest, indicating that these categories are already distinguished quite clearly, leaving limited space for improvement. The visual results in Fig. 6 further show that S2CMamba generates clearer boundaries and more stable predictions.

2) Trento Dataset: Table III presents the quantitative classification results on the Trento dataset. The proposed S2CMamba achieves the best overall performance, with OA, AA, and Kappa reaching 99.09%, 98.49%, and 98.79%, respectively, outperforming all competing methods. Compared with the second-best MS2CANet, S2CMamba improves OA, AA, and Kappa by 0.25%, 0.57%, and 0.34%, respectively, demonstrating its effectiveness for joint HSI and LiDAR classification in high-resolution agricultural scenes. Methods focusing on global feature interaction, such as HyperMLP and IDNet, are effective in modeling long-range dependencies but generally lack explicit spatial structure modeling, which constrains their performance in complex agricultural scenarios. By contrast, S3F2Net combines CNN and GCN to simultaneously capture local spatial details and global topological dependencies, leading to more comprehensive multimodal feature representation. In contrast,

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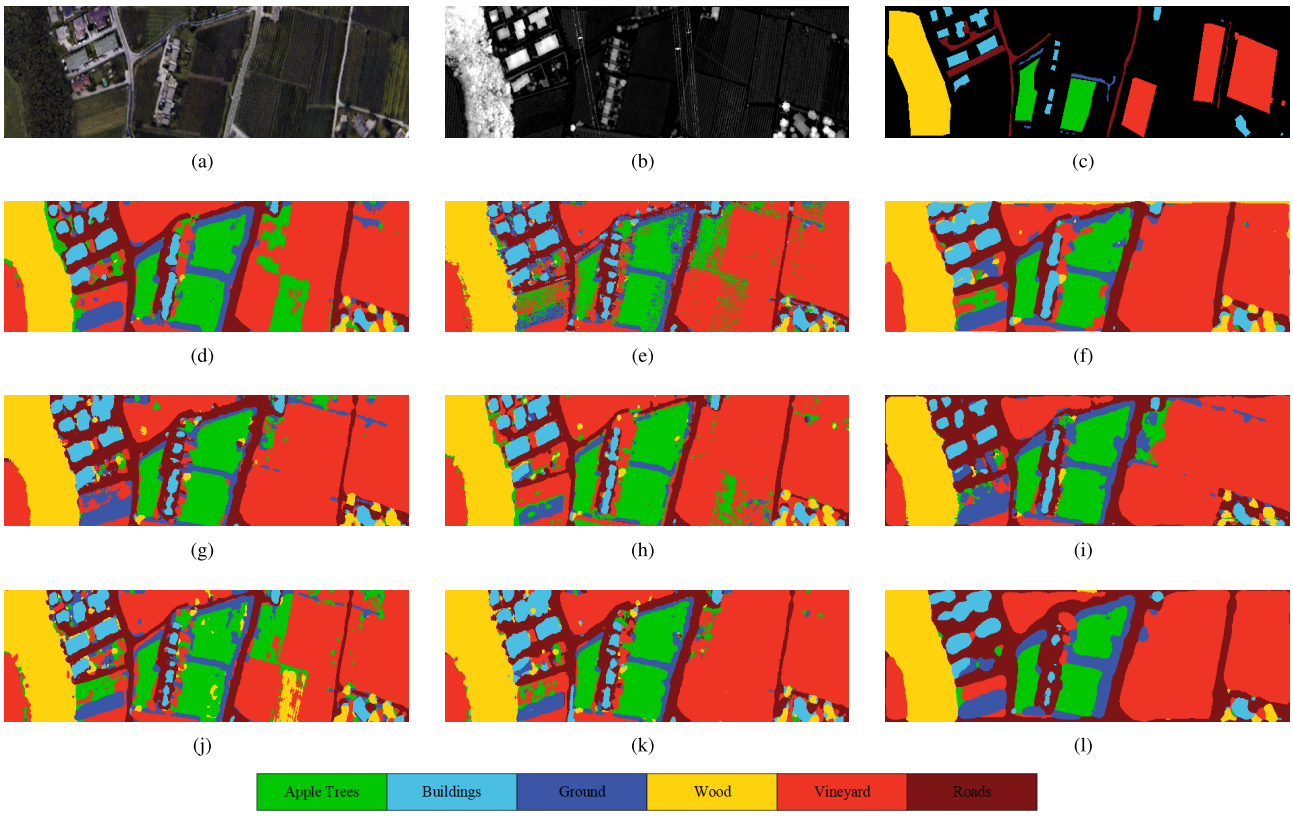


Fig. 7. Classification map of the various methods used in the Trento dataset. (a) Pseudocolor composite image based on bands 20, 15, and 5 of the HSI data. (b) Grayscale image for DSM based on LiDAR data. (c) Ground truth. (d) MICFNet (98.76%). (e) HyperMLP (96.47%). (f) GLTNet (98.63%). (g) MS2CANet (98.84%). (h) IDNet (97.72%). (i) HLMamba (97.59%). (j) E-Mamba (95.93%). (k) S3F2Net (97.93%). (l) Our methods (99.09%).

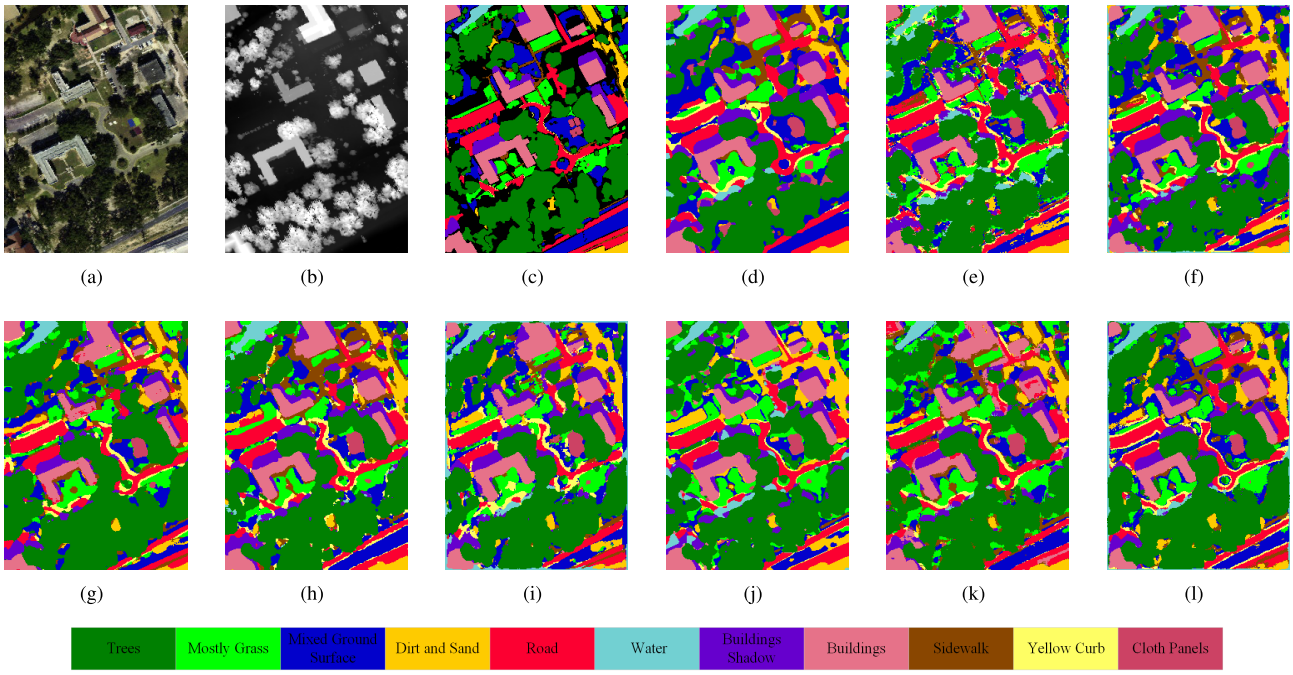


Fig. 8. Classification map of the various methods used in the MUUFL dataset. (a) Pseudocolor composite image based on bands 20, 15, and 5 of the HSI data. (b) Grayscale image for DSM based on LiDAR data. (c) Ground truth. (d) MICFNet (79.86%). (e) HyperMLP (79.98%). (f) GLTNet (84.16%). (g) MS2CANet (81.16%). (h) IDNet (81.08%). (i) HLMamba (81.90%). (j) E-Mamba (79.62%). (k) S3F2Net (83.13%). (l) Our methods (85.26%).

Mamba-based methods, including HLMamba and E-Mamba, mainly rely on sequential state-space modeling and still show limitations in representing complex spatial patterns. The superior performance of S2CMamba mainly benefits from the coordinated effects of S2SSM, SDU, and CGAM. Specifically, S2SSM captures multidirectional spatial structural

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TABLE II
COMPARATIVE CLASSIFICATION PERFORMANCE OF VARIOUS METHODS ON THE HOUSTON DATASET

No.	MICFNet	HyperMLP	GLTNet	MS2CANet	IDNet	HLMamba	E-Mamba	S3F2Net	Our methods
C1	94.47	95.32	93.22	87.09	93.40	91.03	94.18	92.33	**96.10**
C2	89.11	98.01	95.11	95.64	94.46	95.51	**97.62**	89.84	96.57
C3	99.13	99.74	99.04	97.60	99.38	99.85	**99.88**	99.37	99.46
C4	98.54	**99.75**	97.84	99.40	98.91	96.19	97.56	98.26	98.94
C5	99.60	98.14	99.57	99.75	99.96	99.96	99.96	99.98	**99.99**
C6	97.02	97.58	97.76	98.71	97.64	94.54	98.54	92.27	**98.30**
C7	88.19	96.32	94.59	95.64	93.46	97.43	94.44	87.64	**95.79**
C8	85.60	90.32	89.34	85.29	87.39	91.25	89.07	92.41	**90.92**
C9	84.62	87.84	92.26	83.93	82.76	85.97	87.97	67.58	**91.84**
C10	69.66	91.13	90.53	88.49	95.53	96.83	95.84	62.41	**94.96**
C11	96.15	88.91	91.87	97.27	96.82	98.30	93.47	96.59	**97.58**
C12	83.47	84.53	89.43	87.41	91.99	85.47	92.62	33.82	**90.12**
C13	88.56	89.35	96.63	96.51	95.66	97.77	**97.27**	97.36	96.82
C14	**100**	99.82	99.80	99.87	99.93	99.97	99.95	**100**	99.95
C15	99.22	99.98	99.90	99.97	99.97	99.95	**100**	99.95	99.98
OA(%)	90.33±1.53	93.93±0.89	94.29±1.17	93.08±0.80	94.35±0.63	94.64±2.36	95.11±0.71	84.86±2.10	**95.93±0.26**
AA(%)	91.56±1.23	94.56±0.74	95.13±0.89	94.17±0.68	95.15±0.64	95.33±2.23	95.89±0.56	87.32±1.68	**96.46±0.22**
Kappa	89.54±1.65	94.43±0.97	93.82±1.26	92.57±0.85	93.88±0.68	94.20±2.55	94.71±0.76	83.64±2.27	**95.59±0.28**

TABLE III
COMPARATIVE CLASSIFICATION PERFORMANCE OF VARIOUS METHODS ON THE TRENTO DATASET

No.	MICFNet	HyperMLP	GLTNet	MS2CANet	IDNet	HLMamba	E-Mamba	S3F2Net	Our methods
C1	95.55	93.14	98.30	98.88	99.37	98.93	94.91	99.50	**98.65**
C2	95.58	96.26	99.01	96.17	96.15	85.73	92.24	91.96	**99.25**
C3	97.92	94.63	96.33	97.93	95.18	**100**	98.44	95.65	98.71
C4	99.94	99.63	**100**	99.99	99.98	**100**	99.60	99.84	99.98
C5	99.74	96.91	**100**	99.85	96.73	**100**	97.61	99.85	99.82
C6	99.27	90.66	90.53	94.67	94.20	91.49	84.04	89.89	**94.55**
OA(%)	98.76±0.38	96.47±0.96	98.63±0.35	98.84±0.24	97.72±0.59	97.59±0.04	95.93±1.15	97.93±0.47	**99.09±0.13**
AA(%)	98.00±0.46	95.21±0.81	97.36±0.60	97.92±0.37	96.93±0.48	96.02±0.05	94.48±1.89	96.12±0.91	**98.49±0.37**
Kappa	98.35±0.50	95.30±1.27	98.17±0.46	98.45±0.32	96.97±0.78	96.79±0.05	94.57±1.54	97.24±0.63	**98.79±0.17**

TABLE IV
COMPARATIVE CLASSIFICATION PERFORMANCE OF VARIOUS METHODS ON THE MUUFL DATASET

No.	MICFNet	HyperMLP	GLTNet	MS2CANet	IDNet	HLMamba	E-Mamba	S3F2Net	Our methods
C1	83.39	86.54	89.82	**93.37**	86.07	93.18	84.86	89.81	89.85
C2	81.11	72.32	75.88	67.32	73.13	77.68	82.41	75.06	**78.95**
C3	56.02	55.46	65.87	63.48	63.83	61.17	45.10	69.24	**68.62**
C4	77.29	74.55	86.78	88.30	90.86	**92.44**	93.05	90.09	89.25
C5	80.84	80.17	81.92	68.06	80.50	79.42	89.22	80.45	**84.16**
C6	**100**	98.88	99.66	85.72	99.06	98.59	98.57	**99.57**	99.79
C7	84.16	84.89	86.64	70.61	86.84	89.13	77.28	81.50	**90.38**
C8	92.94	92.29	**94.00**	84.87	93.72	95.07	95.82	85.36	92.81
C9	58.69	50.89	59.30	58.13	50.89	48.35	16.42	53.68	**61.76**
C10	73.93	68.85	72.21	76.81	72.15	71.35	32.64	78.28	**79.41**
C11	97.27	92.64	96.27	86.55	97.95	98.80	**99.28**	95.98	96.49
OA(%)	79.86±1.38	79.98±1.86	84.16±1.12	81.16±3.17	81.08±1.71	81.90±0.15	79.62±2.91	83.13±1.13	**85.26±1.42**
AA(%)	80.51±1.26	77.95±1.53	82.52±1.44	76.66±6.22	81.36±1.18	80.32±0.46	74.06±1.39	81.73±1.33	**84.68±0.96**
Kappa	74.40±0.02	74.38±2.18	79.54±1.36	76.01±3.79	76.71±2.09	76.51±0.21	73.97±3.31	78.16±1.33	**80.97±1.72**

dependencies, SDU reduces cross-modal semantic ambiguity, and CGAM enhances foreground–background discrimination, thereby improving the discriminative capability of joint HSI–LiDAR representations. In terms of performance across categories, the most significant improvement was observed in buildings (C2), while steady gains were also achieved in apple trees (C1) and roads (C6). For ground (C3) and vineyard (C5), although the results were close to the best methods, they were slightly lower, which indicates that existing methods are already capable of handling these categories effectively. This advantage is further confirmed by the visualization in Fig. 7, where competing methods tend to produce blurred or fragmented class boundaries, while S2CMamba preserves clearer boundaries for different land-cover categories, verifying the effectiveness of S2SSM and CGAM in structural modeling and foreground–background separation.

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TABLE V
ABLATION STUDY OF THE KEY COMPONENTS OF OUR METHOD (BEST RESULTS IN BOLD)

S2SSM	SDU	CGAM	The overall accuracy of different datasets		
			Houston2013	Trento	MUUFL
×	√	√	94.38	97.91	83.01
√	×	√	95.32	97.67	82.60
√	√	×	93.96	96.04	81.26
√	×	×	94.76	96.11	81.99
√	√	√	**95.93**	**99.09**	**85.26**

TABLE VI
ABLATION STUDY OF THE STRUCTURAL MODELING STRATEGY IN THE PROPOSED S2SSM MODULE (BEST RESULTS IN BOLD)

SS2D	1D Mamba	SAVSSLayer	The overall accuracy of different datasets		
			Houston2013	Trento	MUUFL
√	×	×	94.85	97.94	83.19
×	×	√	94.97	98.75	84.85
×	√	√	93.12	97.73	81.64
√	×	√	**95.93**	**99.09**	**85.26**

3) MUUFL Dataset: Table IV reports the classification results on the MUUFL dataset. S2CMamba achieves the best overall performance, with OA, AA, and Kappa of 85.26%, 84.68%, and 80.97%, respectively, exceeding GLTNet by 1.10%, 2.16%, and 1.43%. The MUUFL dataset is challenging due to dispersed class distributions, irregular boundaries, and severe shadow interference. Lower-performing methods generally rely more on local modeling or insufficiently constrained multimodal fusion, making them prone to noisy predictions and fragmented regions. In contrast, stronger baselines, such as GLTNet, S3F2Net, HLMamba, and IDNet, share a common advantage in modeling broader contextual or topological relationships, which improves classification consistency in complex scenes. Compared with these methods, S2CMamba further improves performance by jointly addressing structural dependency, semantic ambiguity, and foreground–background interference. Specifically, S2SSM captures long-range structural dependencies among scattered objects, SDU alleviates semantic confusion caused by shadows and adjacent classes, and CGAM further refines discriminative feature aggregation. The most remarkable improvements were observed in buildings shadow (C7), sidewalk (C9), and yellow curb (C10), where S2CMamba achieved the highest accuracy, demonstrating its advantage in handling dispersed and irregular regions. However, challenges remain in mixed ground surface (C3) and dirt and sand (C4), where accuracy remains relatively low or is not the highest, which reflects the difficulties caused by spectral ambiguity and complex spatial patterns. As shown in Fig. 8, these designs enable S2CMamba to generate clearer boundaries and more complete class regions, producing classification maps closer to the ground truth.

E. Ablation Experiments

Ablation experiments were conducted to evaluate the efficacy of the key components of S2CMamba, including the S2SSM, SDU, and CGAM modules. In addition, to further analyze the contribution of the structural modeling strategy, ablations were performed on the SS2D mechanism and the combination scheme of the SAVSSLayer.

The proposed SDU module aims to mitigate cross-modal semantic entanglement between HSI and LiDAR data while enhancing the discriminability of multimodal feature representations, and the proposed CGAM module intensifies foreground-background contrast to promote more efficient feature aggregation. To evaluate their effectiveness, we sequentially removed the proposed SDU and CGAM modules and analyzed the resulting performance variations. As shown in Table V, retaining only the proposed S2SSM module leads to varying degrees of performance degradation compared with the complete framework. For example, on the Houston dataset, removing the SDU or CGAM module reduces the OA to 95.32% and 93.96%, respectively. On the MUUFL dataset, the OA drops to 82.60% and 81.26%, respectively. When both SDU and CGAM are simultaneously removed, the performance further decreases to 94.76%, 96.11%, and 81.99% across the evaluated datasets. The results demonstrate that integrating the proposed SDU and CGAM modules enables the model to more effectively leverage the complementary information from HSI and LiDAR modalities. Specifically, the proposed SDU learns more discriminative semantic representations by explicitly decoupling foreground and background semantic responses in the fused HSI–LiDAR features. Meanwhile, the proposed CGAM enhances inter-class separability by emphasizing foreground–background contrast during multimodal feature aggregation.

To further evaluate the efficacy of the proposed S2SSM module, we performed ablation studies on its essential components, including SS2D, 1-D Mamba, and SAVSSLayer, as shown in Table VI. When only SS2D is employed without introducing SAVSSLayer, the performance across all datasets is inferior to that of the complete model. Replacing SS2D with the conventional 1-D Mamba further degrades classification accuracy; for instance, the OA on the Houston dataset decreases to 93.12%. In contrast, the combined use of SS2D and SAVSSLayer results in optimal performance across all datasets. This result demonstrates that the SS2D can effectively capture multidirectional spatial dependencies in HSI and LiDAR data, while SAVSSLayer further strengthens the representation of cross-modal structural information, thereby improving classification performance in complex RS scenarios.

In summary, the proposed S2CMamba framework integrates three key components, namely, S2SSM, SDU, and CGAM, to address the challenges of HSI–LiDAR joint classification. Specifically, S2SSM performs structure-aware spatial

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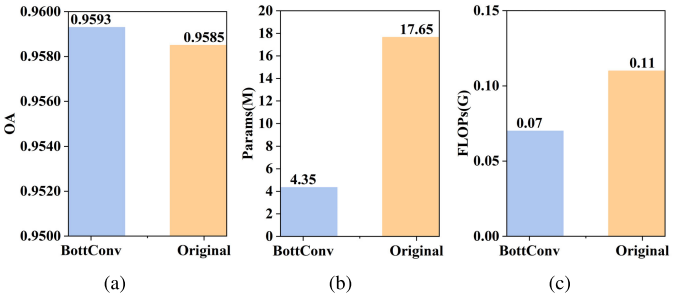


Fig. 9. Performance comparison between BottConv and original convolutions in GBC on the Houston2013 dataset. (a) Comparison of OA Classification Accuracy. (b) Comparison of Parameter Sizes. (c) Comparison of FLOPs Performance.

modeling in the HSI and LiDAR branches to encode multidirectional spatial structures and capture long-range global dependencies. SDU further enhances feature discrimination by explicitly decoupling foreground and background semantic responses in the fused HSI–LiDAR representations. Building upon these semantic priors, CGAM performs contrast-modulated local aggregation to improve intraclass compactness and class-boundary separability in the multimodal feature space. The integration of these components enables the proposed framework to achieve more stable and robust classification performance. The ablation results clearly indicate that each proposed component beneficially impacts the ultimate performance, highlighting the novelty and effectiveness of the S2CMamba framework for multimodal HSI–LiDAR representation learning.

Furthermore, to evaluate the computational efficiency of the proposed framework, additional experiments were conducted to validate the effectiveness of the proposed lightweight GBC module, as shown in Fig. 9. As illustrated in Fig. 9(a), replacing the original convolution with BottConv maintains a comparable OA of 95.93% on the Houston dataset. Meanwhile, Fig. 9(b) and (c) show that the model complexity is significantly reduced, with the number of parameters decreasing from 17.65 to 4.35 M and FLOPs from 0.11 to 0.07 G. These results demonstrate that integrating the proposed GBC module into the S2SSM framework for HSI–LiDAR data achieves an effective balance between classification accuracy and computational efficiency.

V. Conclusion

This article proposes a novel S2CMamba framework for the joint classification of HSI and LiDAR data. First, the S2SSM is designed to capture multidirectional spatial dependencies and long-range contextual information, enhancing the modeling of complex spatial structures in RS scenes. Then, the SDU is used to explicitly decouple foreground and background semantics, effectively suppressing background interference and alleviating semantic ambiguity during feature fusion. Second, the CGAM is developed to perform contrast-guided local feature aggregation, which enhances intraclass compactness and improves class-boundary discrimination. Experimental results using various benchmark datasets demonstrate the superiority of S2CMamba in the joint classification of HSI and LiDAR data. Future work will focus on extending the framework to broader multimodal RS scenarios.

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